

7	(a) wavelength of wave associated with a particle that is moving	M1 A1	[2]
	(b) (i) energy of electron = $850 \times 1.6 \times 10^{-19}$ = 1.36×10^{-16} J energy = $p^2 / 2m$ or $p = mv$ <u>and</u> $E_K = \frac{1}{2}mv^2$ momentum = $\sqrt{(1.36 \times 10^{-16} \times 2 \times 9.11 \times 10^{-31})}$ = 1.6×10^{-23} N s	M1 M1 A0	[2]
	(ii) $\lambda = h / p$ wavelength = $(6.63 \times 10^{-34}) / (1.6 \times 10^{-23})$ = 4.1×10^{-11} m	C1 A1	[2]
	(c) diagram or description showing: electron beam in a vacuum incident on <u>thin</u> metal target / carbon <u>film</u> fluorescent screen pattern of concentric rings observed pattern similar to diffraction pattern observed with visible light	B1 B1 B1 M1 A1	[5]